

Lab 4: Data Visualization and EDA

CPE232 Data Models

1. Load all Superstore datasets.

Note: The same datasets used in Lab 3

```
In [36]: # Write your code here
import pandas as pd

superstore_order = pd.read_csv('Superstore/superstore_order.csv')
superstore_people = pd.read_csv('Superstore/superstore_people.csv')
superstore_return = pd.read_csv('Superstore/superstore_return.csv')
```

2. Determine shape of each dataset (print out the results as well).

```
In [37]: # Write your code here
print(f"Order: {superstore_order.shape}" )
print(f"People: {superstore_people.shape}" )
print(f"Return: {superstore_return.shape}")
```

```
Order: (8880, 21)
People: (4, 2)
Return: (296, 2)
```

3. Show information of the dataset.

```
In [38]: # Write your code here
print(superstore_order.info())
```

```

<class 'pandas.DataFrame'>
RangeIndex: 8880 entries, 0 to 8879
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row ID                 8880 non-null  int64
1   Order ID               8880 non-null  str
2   Order Date             8880 non-null  str
3   Ship Date              8880 non-null  str
4   Ship Mode              8880 non-null  str
5   Customer ID            8880 non-null  str
6   Customer Name          8880 non-null  str
7   Segment                8880 non-null  str
8   Country                8880 non-null  str
9   City                   8880 non-null  str
10  State                  8880 non-null  str
11  Postal Code            8880 non-null  int64
12  Region                 8880 non-null  str
13  Product ID             8880 non-null  str
14  Category               8880 non-null  str
15  Sub-Category           8880 non-null  str
16  Product Name           8880 non-null  str
17  Sales                  8880 non-null  float64
18  Quantity               8880 non-null  int64
19  Discount               8880 non-null  float64
20  Profit                 8880 non-null  float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.4 MB
None

```

```
In [39]: print(superstore_people.info())
```

```

<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 2 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Person  4 non-null        str
1   Region  4 non-null        str
dtypes: str(2)
memory usage: 196.0 bytes
None

```

```
In [40]: print(superstore_return.info())
```

```

<class 'pandas.DataFrame'>
RangeIndex: 296 entries, 0 to 295
Data columns (total 2 columns):
#   Column    Non-Null Count  Dtype
---  -
0   Returned  296 non-null    str
1   Order ID  296 non-null    str
dtypes: str(2)
memory usage: 4.8 KB
None

```

4. Are there any missing values? If so, in which column?

Ans: No, there aren't.

```
In [41]: print(superstore_order.isnull().sum())
```

```
Row ID      0
Order ID    0
Order Date  0
Ship Date   0
Ship Mode   0
Customer ID  0
Customer Name 0
Segment     0
Country     0
City        0
State       0
Postal Code  0
Region      0
Product ID  0
Category    0
Sub-Category 0
Product Name 0
Sales       0
Quantity    0
Discount    0
Profit      0
dtype: int64
```

```
In [42]: print(superstore_people.isnull().sum())
```

```
Person      0
Region      0
dtype: int64
```

```
In [43]: print(superstore_return.isnull().sum())
```

```
Returned    0
Order ID    0
dtype: int64
```

5.

- 5.1 List unique segments
- 5.2 List unique segments and their corresponding count
- 5.3 Create a pie chart to demonstrate unique segments and their count
- 5.4 Briefly describe what could be interpreted from this pie chart

Note: please create additional cells to answer 5.2 - 5.3

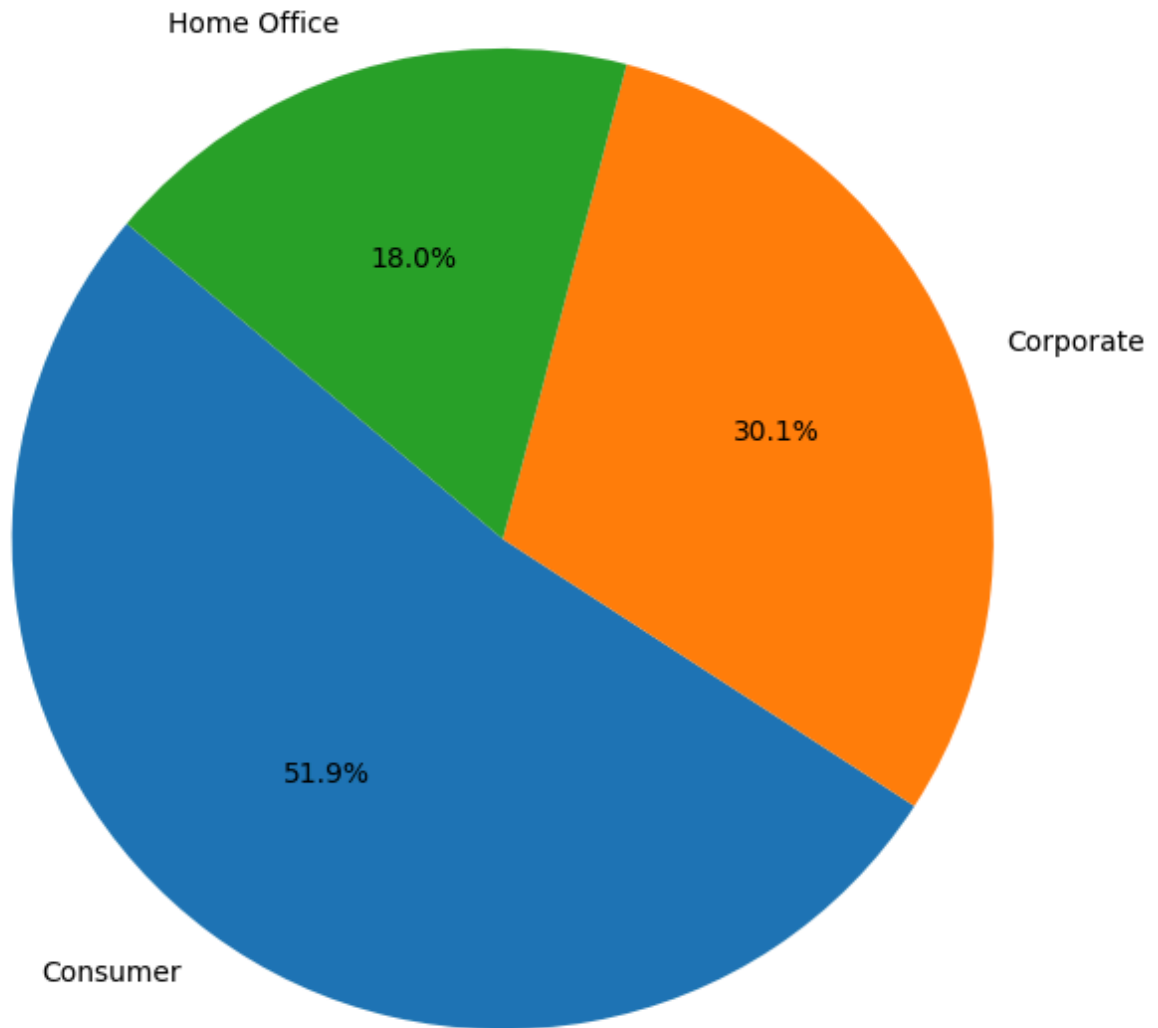
```
In [44]: # Write your code here (5.1)
unique_seg = superstore_order['Segment'].unique()
print(unique_seg)
```

```
<StringArray>  
['Consumer', 'Corporate', 'Home Office']  
Length: 3, dtype: str
```

```
In [ ]: #5.2  
seg_counts = superstore_order['Segment'].value_counts()  
print(seg_counts)
```

```
Segment  
Consumer      4613  
Corporate     2673  
Home Office   1594  
Name: count, dtype: int64
```

```
In [ ]: #5.3  
import matplotlib.pyplot as plt  
  
plt.figure(figsize=(7, 7))  
plt.pie(  
    seg_counts,  
    labels=seg_counts.index,  
    autopct='%1.1f%%',  
    startangle=140,  
)  
  
plt.axis('equal')  
plt.show()
```



Answer for the question 5.4

Ans: From pie chart, this business has highly focus on Consumer, accounting for 51.9% of the total and the Corporate segment makes up the second-largest portion at 30.1%, while the Home Office segment accounts for the remaining 18.0%.

6.

- 6.1 List unique states
- 6.2 List top-10 unique states and their corresponding count
- 6.3 Create a bar chart (vertical) to demonstrate the count of top-10 unique states
- 6.4 Based on 6.2, also include the total sales of these states (show your result as a dataframe)
- 6.5 Using the result from 6.4, if you were the owner of this superstore, what information could be interpreted from this result?

Note: please create additional cells to answer 6.2 - 6.4

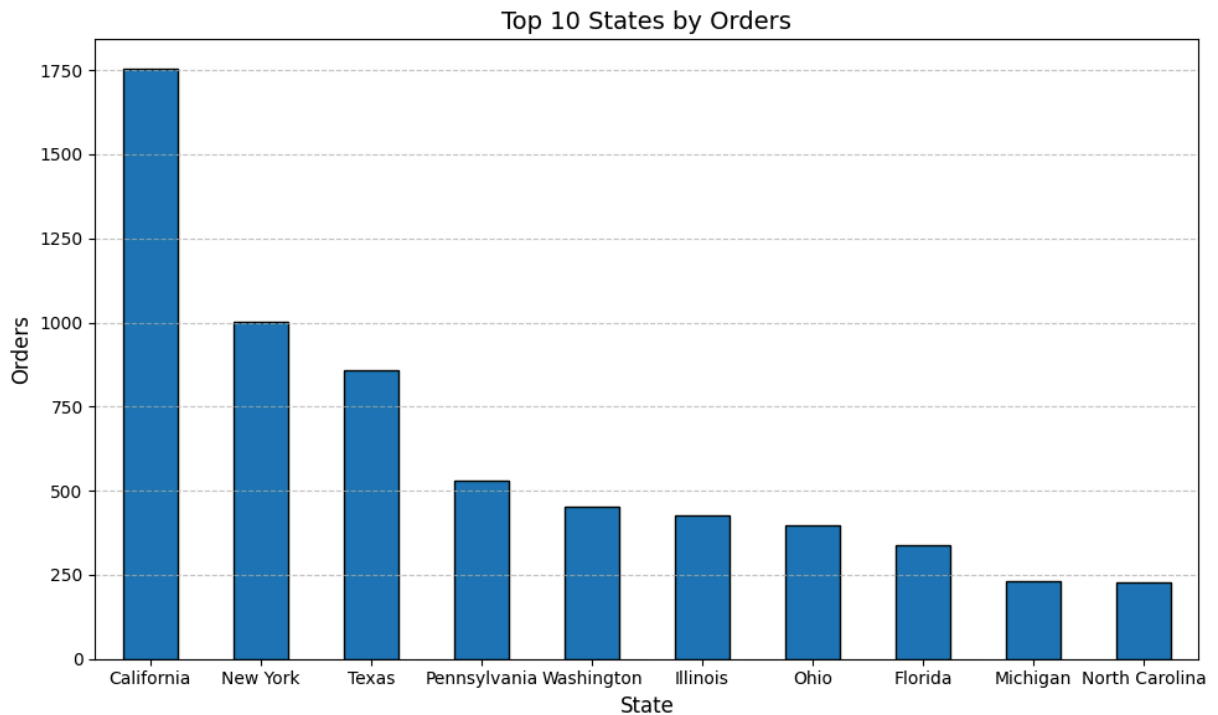
```
In [47]: # Write your code here (6.1)
unique_states = superstore_order['State'].unique()
print(unique_states)
```

```
<StringArray>
[
    'Kentucky',      'California',      'Florida',
    'North Carolina', 'Washington',      'Texas',
    'Wisconsin',      'Utah',            'Nebraska',
    'Pennsylvania',   'Illinois',        'Minnesota',
    'Michigan',        'Delaware',        'Indiana',
    'New York',        'Arizona',          'Virginia',
    'Tennessee',      'Alabama',          'South Carolina',
    'Oregon',          'Colorado',         'Iowa',
    'Ohio',            'Missouri',         'Oklahoma',
    'New Mexico',      'Louisiana',        'Connecticut',
    'New Jersey',      'Massachusetts',    'Georgia',
    'Nevada',          'Rhode Island',     'Mississippi',
    'Arkansas',        'Montana',          'New Hampshire',
    'Maryland', 'District of Columbia', 'Kansas',
    'Vermont',         'Maine',            'South Dakota',
    'Idaho',           'North Dakota',     'Wyoming',
    'West Virginia']
Length: 49, dtype: str
```

```
In [ ]: #6.2
state_top10 = superstore_order['State'].value_counts().nlargest(10)
state_top10
```

```
Out[ ]: State
California      1754
New York        1001
Texas           860
Pennsylvania    531
Washington      452
Illinois        427
Ohio            396
Florida         339
Michigan        230
North Carolina  229
Name: count, dtype: int64
```

```
In [ ]: #6.3
plt.figure(figsize=(10, 6))
state_top10.plot(kind='bar', edgecolor='black')
plt.title('Top 10 States by Orders', fontsize=14)
plt.xlabel('State', fontsize=12)
plt.ylabel('Orders', fontsize=12)
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



```
In [ ]: #6.4
states_top10_list = superstore_order['State'].value_counts().nlargest(10).index.tolist()
top10df = superstore_order[superstore_order['State'].isin(states_top10_list)]
top10_sale = top10df.groupby('State').agg(
    Order=('State', 'count'),
    Total_Sales=('Sales', 'sum')
)
top10_sale = top10_sale.sort_values(by='Order', ascending=False)
print(top10_sale)
```

State	Order	Total_Sales
California	1754	399195.4555
New York	1001	274866.8190
Texas	860	147855.0282
Pennsylvania	531	103852.5210
Washington	452	124497.7780
Illinois	427	71456.1780
Ohio	396	67924.2140
Florida	339	84083.0880
Michigan	230	62147.6960
North Carolina	229	49962.1580

Answer for the question 6.5

Ans: From the table, this supermarket has mainly sales in California, New York and Texas by comparing the other top 10 sales that have equal or less than 600 order.

7.

- 7.1 List unique categories
- 7.2 Create a bar chart (horizontal) to demonstrate the proportion of these categories

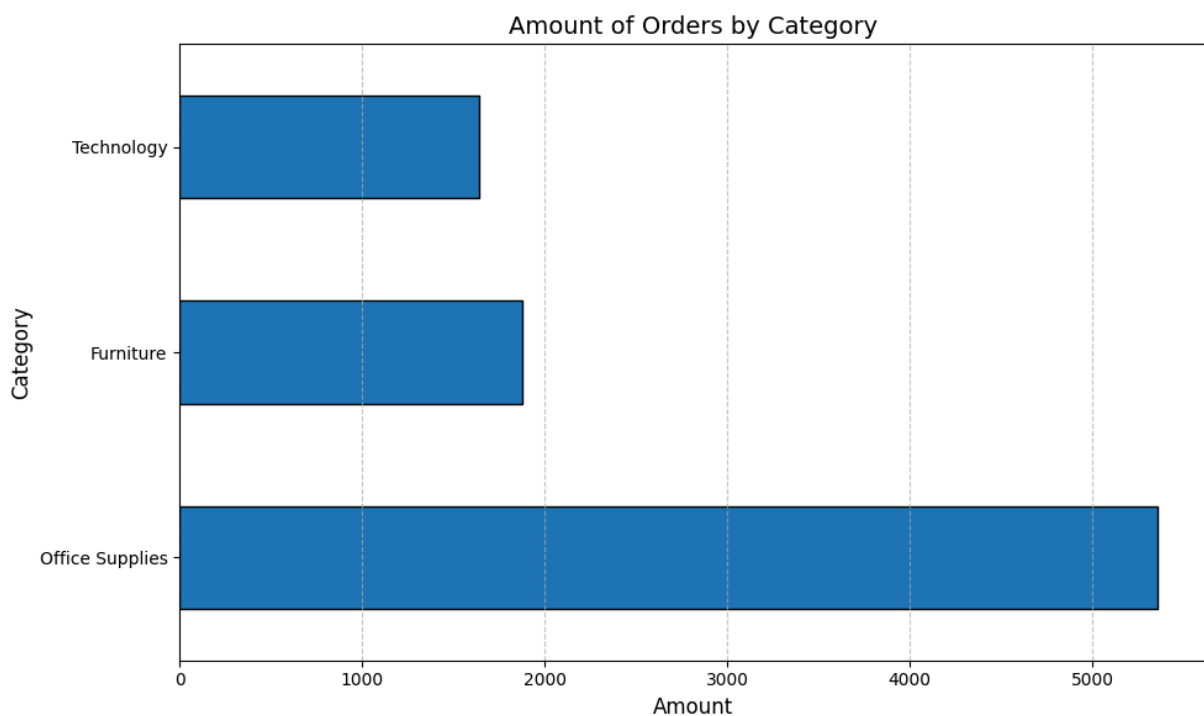
- 7.3 Compute the ratio of these categories in percentage and print the results

Note: please create additional cells to answer 7.2 - 7.3

```
In [50]: # Write your code here (7.1)
unique_category = superstore_order['Category'].unique()
print(unique_category)
```

```
<StringArray>
['Furniture', 'Office Supplies', 'Technology']
Length: 3, dtype: str
```

```
In [ ]: #7.2
category_count = superstore_order['Category'].value_counts()
plt.figure(figsize=(10, 6))
category_count.plot(kind='barh', edgecolor='black')
plt.title('Amount of Orders by Category', fontsize=14)
plt.xlabel('Amount', fontsize=12)
plt.ylabel('Category', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



```
In [ ]: #7.3
category_percentages = superstore_order['Category'].value_counts(normalize=True) * 100
print("Percentage of Orders by Category:")
print(category_percentages)
```


Percentage of Orders by Category:

Category

Office Supplies 60.360360

Furniture 21.171171

Technology 18.468468

Name: proportion, dtype: float64

8. Update the type of all columns that contain dates to *datetime* and show information after an update.

In [53]: # write your code here

```
superstore_order['Order Date'] = pd.to_datetime(superstore_order['Order Date'], format='%m/%d/%Y')
superstore_order['Ship Date'] = pd.to_datetime(superstore_order['Ship Date'], format='%m/%d/%Y')
superstore_order.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8880 entries, 0 to 8879
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Row ID              8880 non-null   int64
1   Order ID            8880 non-null   str
2   Order Date          8880 non-null   datetime64[us]
3   Ship Date           8880 non-null   datetime64[us]
4   Ship Mode           8880 non-null   str
5   Customer ID         8880 non-null   str
6   Customer Name       8880 non-null   str
7   Segment             8880 non-null   str
8   Country             8880 non-null   str
9   City                8880 non-null   str
10  State               8880 non-null   str
11  Postal Code         8880 non-null   int64
12  Region              8880 non-null   str
13  Product ID          8880 non-null   str
14  Category            8880 non-null   str
15  Sub-Category        8880 non-null   str
16  Product Name        8880 non-null   str
17  Sales               8880 non-null   float64
18  Quantity            8880 non-null   int64
19  Discount            8880 non-null   float64
20  Profit              8880 non-null   float64
dtypes: datetime64[us](2), float64(3), int64(3), str(13)
memory usage: 1.4 MB
```

9. Create a new column "Processing time day" to show number of days taken to ship an order and show your result in a dataframe format.

Hint: The duration starts as soon as the item has been ordered and ends once the order has successfully shipped.

In [54]: # write your code here

```
superstore_order['Processing time day'] = (superstore_order['Ship Date'] - superstore_order['Order Date']).dt.days
```

```
print(superstore_order[['Order Date', 'Ship Date', 'Processing time day']].head())
```

	Order Date	Ship Date	Processing time day
0	2016-11-08	2016-11-11	3
1	2016-11-08	2016-11-11	3
2	2016-06-12	2016-06-16	4
3	2015-10-11	2015-10-18	7
4	2015-10-11	2015-10-18	7

10. Based on the result in 9.

- 10.1 How many orders are there that take more than 5 days to process?
- 10.2 Show the top 5 rows (expected output should contain these columns: Order ID, Order Date, Ship Date, Processing time day, Quantity)
- 10.3 Plot the histogram based on the column Quantity

Note: please create additional cells to answer 10.2 - 10.3

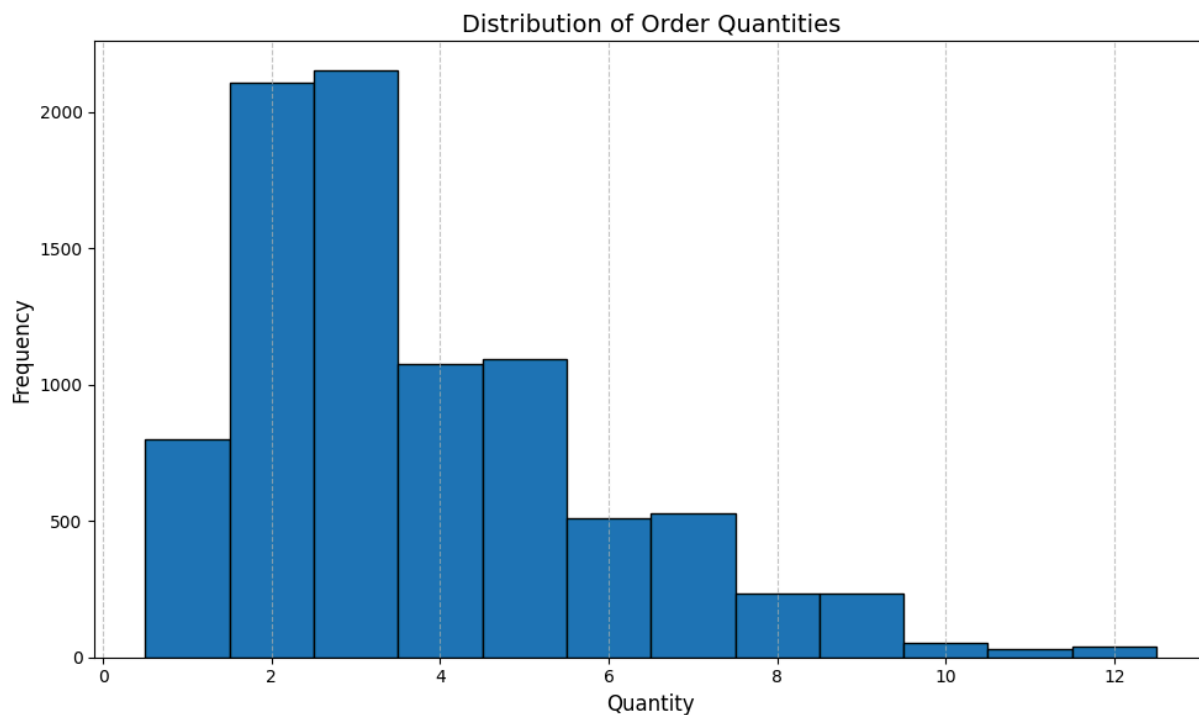
```
In [55]: # Write your code here (10.1)
ordermorethan5 = superstore_order[superstore_order['Processing time day'] > 5]
print(len(ordermorethan5))
```

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```
In [ ]: #10.2
ordermorethan5 = ordermorethan5.sort_values(by='Processing time day', ascending=False)
print(ordermorethan5[['Order ID', 'Order Date', 'Ship Date', 'Processing time day',
```

	Order ID	Order Date	Ship Date	Processing time day	Quantity
86	CA-2017-155558	2017-10-26	2017-11-02	7	1
87	CA-2017-155558	2017-10-26	2017-11-02	7	2
3	US-2015-108966	2015-10-11	2015-10-18	7	5
8860	CA-2015-125934	2015-09-11	2015-09-18	7	3
8861	CA-2015-125934	2015-09-11	2015-09-18	7	4

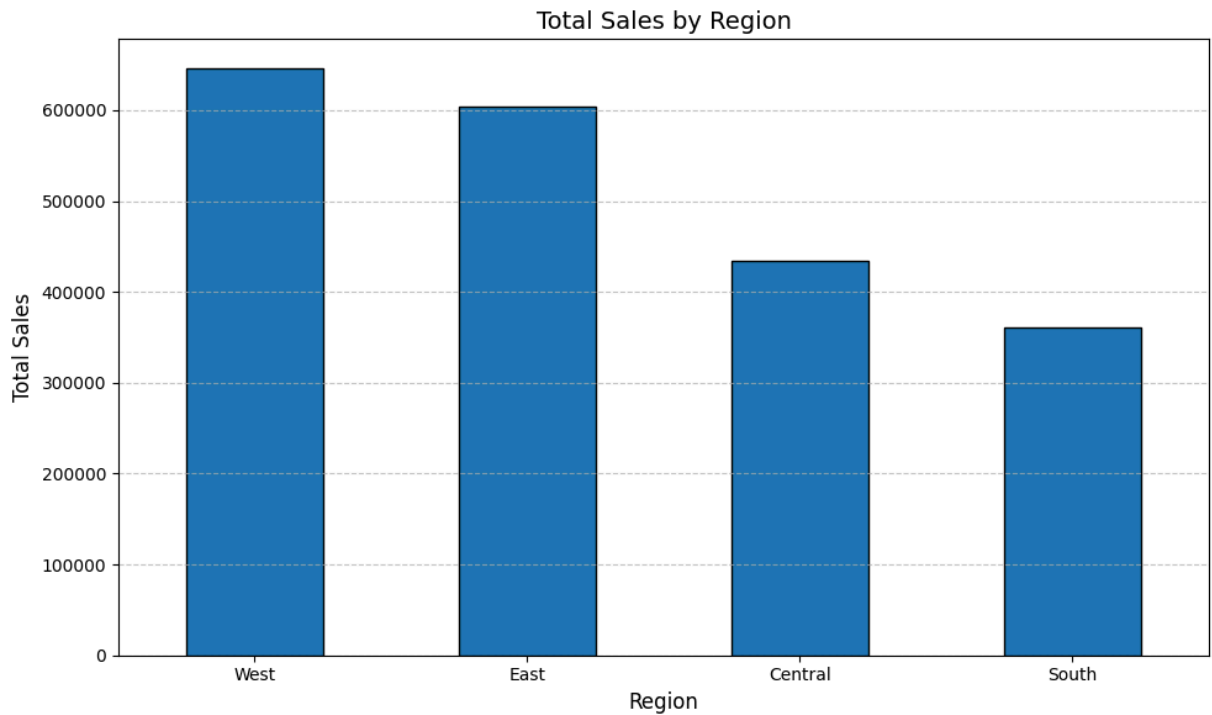
```
In [ ]: #10.3
plt.figure(figsize=(10, 6))
plt.hist(superstore_order['Quantity'],
         bins=range(1, superstore_order['Quantity'].max()),
         edgecolor='black',
         align='left')
plt.title('Distribution of Order Quantities', fontsize=14)
plt.xlabel('Quantity', fontsize=12)
plt.ylabel('Frequency', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



11. Total sales compare across different regions

- 11.1 Create a bar chart to visualize.

```
In [ ]: # Write your code here (11.1)
salebyregion = superstore_order.groupby('Region')['Sales'].sum().sort_values(ascending=True)
plt.figure(figsize=(10, 6))
salebyregion.plot(kind='bar', edgecolor='black')
plt.title('Total Sales by Region', fontsize=14)
plt.xlabel('Region', fontsize=12)
plt.ylabel('Total Sales', fontsize=12)
plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



In [154... `salebyregion`

Out[154... `Region`
 West 645943.7905
 East 603648.0590
 Central 434723.2222
 South 361376.7910
 Name: Sales, dtype: float64

- 11.2 How do total sales compare across different regions? Explain in as much detail as possible.

Ans: From the chart, The main market regions are West and East. These two are the primary revenue for the company, significantly better than the other two. The Central region is in a distant third place, with sales hovering around 500,000. The South has the lowest performance of all, generating nearly half the revenue of the top region.

12. Which states have the highest number of returns? Use a horizontal bar chart.

Ans: California has highest number of returns which is 298 times.

```
In [90]: # Write your code here (12)
combined_orderreturn = pd.merge(superstore_order, superstore_return, on='Order ID',
orderreturntop5 = combined_orderreturn['State'].value_counts().nlargest(5)
plt.figure(figsize=(10, 6))
orderreturntop5.plot(kind='bar', edgecolor='black')
plt.title('Top 10 States by Number of Returns', fontsize=14)
plt.xlabel('Number of Returned Items', fontsize=12)
plt.ylabel('States', fontsize=12)
plt.xticks(rotation=0)
```

```
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
print(orderreturntop5)
```

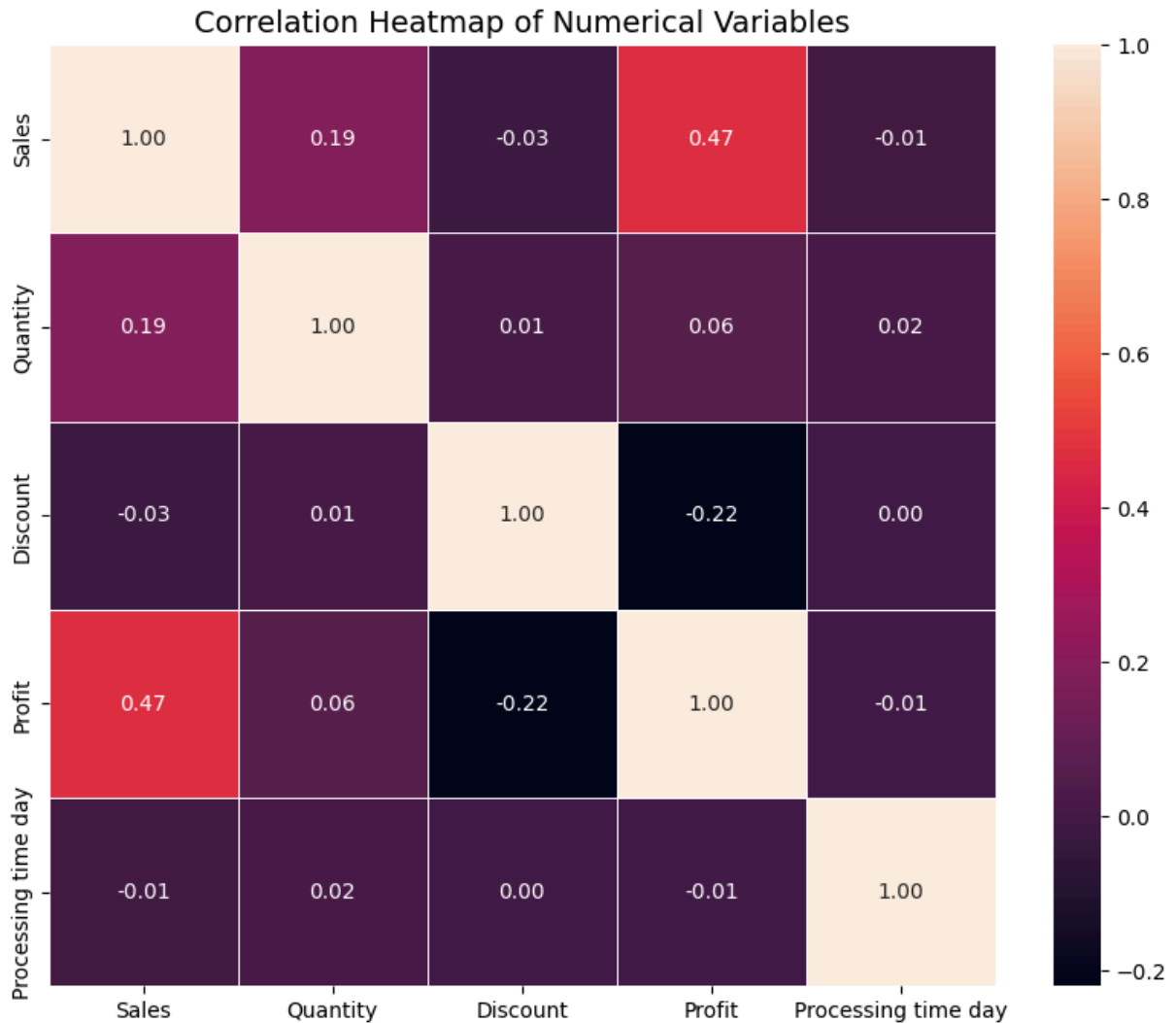


```
State
California    298
Washington     60
New York       59
Texas          30
Colorado       25
Name: count, dtype: int64
```

13. What is the correlation between numerical variables in the superstore_order dataset?
Use a heatmap *Hint: Use seaborn to create a heatmap :*

```
In [96]: # Write your code here (13)
import seaborn as sns
numerical_cols = ['Sales', 'Quantity', 'Discount', 'Profit', 'Processing time day']
corr_data = superstore_order[numerical_cols]
correlation_matrix = corr_data.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix,
            annot=True,
            fmt=".2f",
            linewidths=0.5,)

plt.title('Correlation Heatmap of Numerical Variables', fontsize=14)
plt.show()
```



14. Create a USA State-Level Choropleth Map to visualize total sales per state.

- The darkest color represents the highest total sales.
- The lightest color represents the lowest total sales.
- Use a continuous gradient scale (e.g., dark blue to light blue, dark red to light red, or any custom gradient of your choice).

Hint: Use plotly.express

In [157...

```
# Write your code here (14)
import plotly.express as px
state_sales = superstore_order.groupby('State')['Sales'].sum().reset_index()
state_codes = {
    'Alabama': 'AL', 'Alaska': 'AK', 'Arizona': 'AZ', 'Arkansas': 'AR', 'California': 'CA',
    'Colorado': 'CO', 'Connecticut': 'CT', 'Delaware': 'DE', 'District of Columbia': 'DC',
    'Florida': 'FL', 'Georgia': 'GA', 'Hawaii': 'HI', 'Idaho': 'ID', 'Illinois': 'IL',
    'Indiana': 'IN', 'Iowa': 'IA', 'Kansas': 'KS', 'Kentucky': 'KY', 'Louisiana': 'LA',
    'Maine': 'ME', 'Maryland': 'MD', 'Massachusetts': 'MA', 'Michigan': 'MI', 'Minnesota': 'MN',
    'Mississippi': 'MS', 'Missouri': 'MO', 'Montana': 'MT', 'Nebraska': 'NE', 'Nevada': 'NV',
    'New Hampshire': 'NH', 'New Jersey': 'NJ', 'New Mexico': 'NM', 'New York': 'NY',
    'North Carolina': 'NC', 'North Dakota': 'ND', 'Ohio': 'OH', 'Oklahoma': 'OK', 'Oregon': 'OR',
    'Pennsylvania': 'PA', 'Rhode Island': 'RI', 'South Carolina': 'SC', 'South Dakota': 'SD',
    'Tennessee': 'TN', 'Texas': 'TX', 'Utah': 'UT', 'Vermont': 'VT', 'Virginia': 'VA',
    'Washington': 'WA', 'West Virginia': 'WV', 'Wisconsin': 'WI', 'Wyoming': 'WY'}
```

```

    'Pennsylvania': 'PA', 'Rhode Island': 'RI', 'South Carolina': 'SC', 'South Dakota': 'SD',
    'Tennessee': 'TN', 'Texas': 'TX', 'Utah': 'UT', 'Vermont': 'VT', 'Virginia': 'VA',
    'Washington': 'WA', 'West Virginia': 'WV', 'Wisconsin': 'WI', 'Wyoming': 'WY'
}
state_sales['Code'] = state_sales['State'].map(state_codes)
fig = px.choropleth(
    state_sales,
    locations='Code',
    locationmode="USA-states",
    color='Sales',
    scope="usa",
    color_continuous_scale="Oranges",
    title='Total Sales by State (USA)',
)
fig.show()

```

14.2 Answer the following questions:

1. Which state has the highest total sales?
2. How do sales anomalies affect the gradient color shading on the map?
3. If you change the color scale, does it impact readability? Why or why not?

Ans:

1. California has highest total sales.
2. If a state has a huge outlier compared to others, the map becomes very hard to read.
3. Yes, it does because if you change the color to low contrast you will find it hard to see the different shades.

15. Create a box plot to compare the different shipping modes based on total profit.

```

In [ ]: #Write your code here (15)
plt.figure(figsize=(12, 6))
superstore_order.boxplot(column='Sales', by='Ship Mode', figsize=(12, 7), patch_artist=True)
plt.title('Profit Distribution by Shipping Mode', fontsize=14)
plt.suptitle('')
plt.xlabel('Shipping Mode', fontsize=12)
plt.ylabel('Profit', fontsize=12)

plt.ylim(-200, 400)

plt.show()

```

<Figure size 1200x600 with 0 Axes>



15.2 Which shipping mode has the highest median profit?

Ans: Second Class has highest median profit which is 9.76 dollars

```
In [ ]: median_profits = superstore_order.groupby('Ship Mode')['Profit'].median().sort_values()
print(median_profits)
```

```
Ship Mode
Second Class    9.76080
First Class     8.79255
Same Day        8.43490
Standard Class  8.31040
Name: Profit, dtype: float64
```

[BONUS 20 pts] Determine the percentage of customers who:

- B1) returned the product once
- B2) returned the product at least once
- B3) never returned the product
- Finally, Plot a comparison of B2 and B3

Note: please create additional cells to answer the above points

```
In [ ]: # Write your code here
#B1
data = pd.merge(superstore_order, superstore_return, on='Order ID', how='inner')
return_counts = data.groupby('Customer ID')['Order ID'].nunique()
total_customers = superstore_order['Customer ID'].nunique()
b1_count = (return_counts == 1).sum()
b1_pct = (b1_count/total_customers)*100
print(f"B1: {b1_pct:.2f}% ({b1_count})")
```


B1: 23.57% (186)

```
In [ ]: #B2
b2_count = return_counts.sum()
b2_pct = (b2_count/total_customers)*100
print(f"B2: {b2_pct:.2f}% ({b2_count})")
```

B2: 33.33% (263)

```
In [ ]: #B3
b3_count = total_customers - b2_count
b3_pct = (b3_count/total_customers)*100
print(f"B3: {b3_pct:.2f}% ({b3_count})")
```

B3: 66.67% (526)

```
In [ ]: #Plot a comparison of B2 and B3
labels = ['B2', 'B3']
values = [b2_count, b3_count]
plt.figure(figsize=(8, 6))
plt.pie(values, labels=labels, autopct='%1.1f%%', startangle=90)
plt.title('Customer Return Behavior: B2 vs B3')
plt.axis('equal')
plt.show()
```

Customer Return Behavior: B2 vs B3

